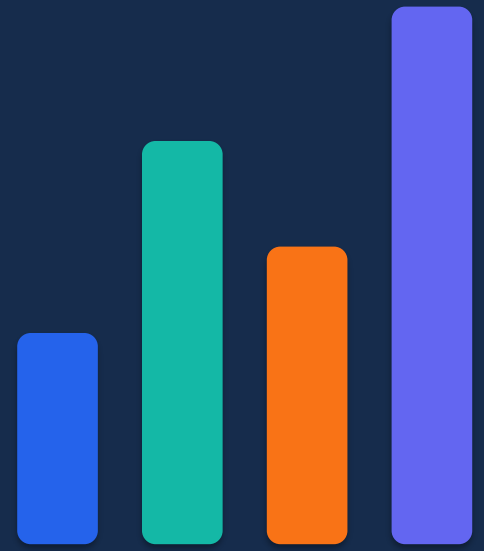
DATA ANALYST PORTFOLIO PROJECT

# Amazon Sales Analysis

SQL-driven business insights across revenue, products, geography and operations

TOOLS • SQL • Python • Data Visualization • PBusiness Recommendations



BUSINESS INTELLIGENCE

# Project overview

## Business objective

Turn transactional sales data into clear decisions about revenue growth, best-selling products, regional performance and fulfilment.

## Dataset

Amazon India order data: 128,938 order rows (120,346 orders) from 31 Mar to 29 Jun 2022.

## Questions covered

- Revenue trend and month-over-month change
- B2B vs B2C order value and basket size
- Best category/size and top 10 SKUs
- City-wise sales leaders
- Cancellation and return patterns
- Fulfilment: Amazon vs Merchant outcomes
- Colour-wise sales performance

**Data & interpretation notes**

Unless a query filters Status, sales amounts include the Amount of cancelled orders (about 8.8% of the total), so they run higher than net revenue. March has only one day of data (31 Mar), so trends are read from April to June. Return status is recorded only for Merchant-fulfilled orders, so return rates are not comparable across fulfilment types.

# 01 | Monthly revenue trend

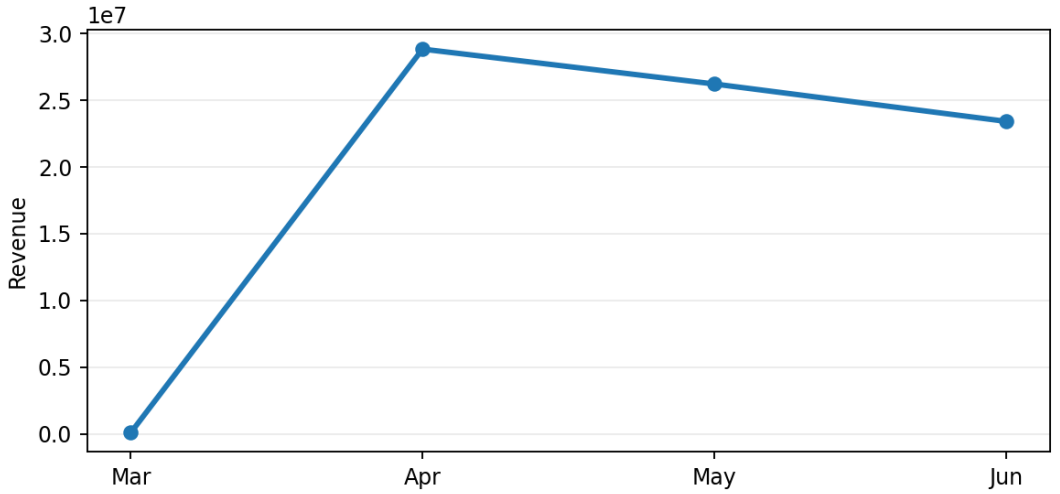
### SQL query

```sql
select month_name,
       month_nu,
       total_sum,
       lag(total_sum) over (order by year, month_nu) as previous,
       round((total_sum - lag(total_sum) over (order by year, month_nu))
             / lag(total_sum) over (order by year, month_nu) * 100, 2) as mom_growth_pct
from (
    select monthname(Date) as month_name,
           month(Date) as month_nu,
           year(Date) as year,
           round(sum(Amount),1) as total_sum
    from amazon
    group by monthname(Date), month(Date), year(Date)
) t
order by year, month_nu;
```

### Query output

| sales_year | month_num | month_name | total_sum | prev_month_sum | mom_growth_pct |
|---|---|---|---|---|---|
| 2022 | 3 | March | 101683.9 | NULL | NULL |
| 2022 | 4 | April | 28830763.3 | 101683.9 | 28253.3 |
| 2022 | 5 | May | 26218621.8 | 28830763.3 | -9.1 |
| 2022 | 6 | June | 23419916.4 | 26218621.8 | -10.7 |

### Revenue by month (visible output)




Insight: Revenue peaks in April, then declines in May (-9.1%) and June (-10.7%). March has only one day of data (31 Mar), so its low bar is not a real dip.

# 02 | B2B vs B2C: order value and basket size

## SQL query

```sql
WITH orders AS (
    SELECT order_id, b2b,
           SUM(amount) AS order_value,
           SUM(qty)    AS items
    FROM amazon
    WHERE status <> 'Cancelled'
    GROUP BY order_id, b2b
)
SELECT b2b,
       COUNT(*)                AS orders,
       ROUND(AVG(order_value)) AS avg_value,
       ROUND(AVG(items), 2)    AS avg_items
FROM orders
GROUP BY b2b;
```

## Query output

| b2b | orders | avg_value | avg_items |
|---|---|---|---|
| 0 | 102440 | 695 | 1.08 |
| 1 | 728 | 768 | 1.12 |

## B2B vs B2C comparison




Insight: B2B orders average ₹768 vs ₹695 for B2C, with a slightly larger basket (1.12 vs 1.08 items). Only 728 B2B orders (0.7%), so treat the gap as indicative.

# 03 | Best-performing category & size

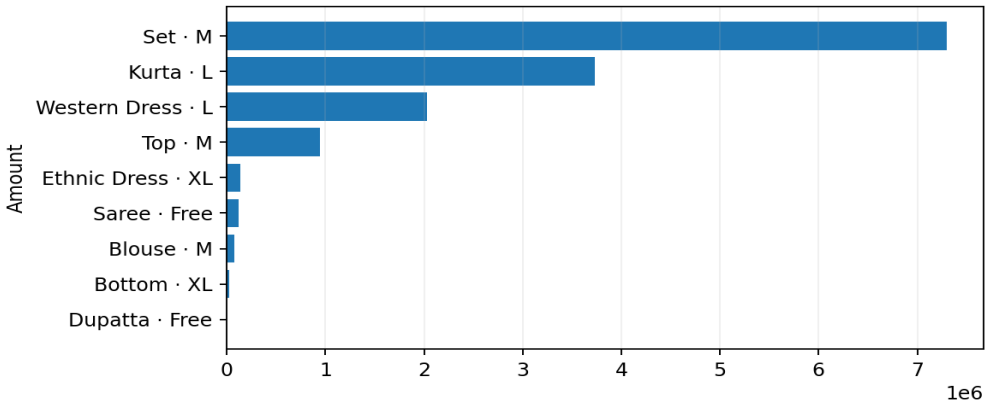
## SQL query

```sql
select category, size, total_amount, rnk
from (
    select category, size, total_amount,
           row_number() over (partition by category order by total_amount desc) as rnk
    from (
        select category, size, sum(amount) as total_amount
        from amazon
        group by category, size
    ) a
) b
where rnk = 1 order by total_amount desc;
```

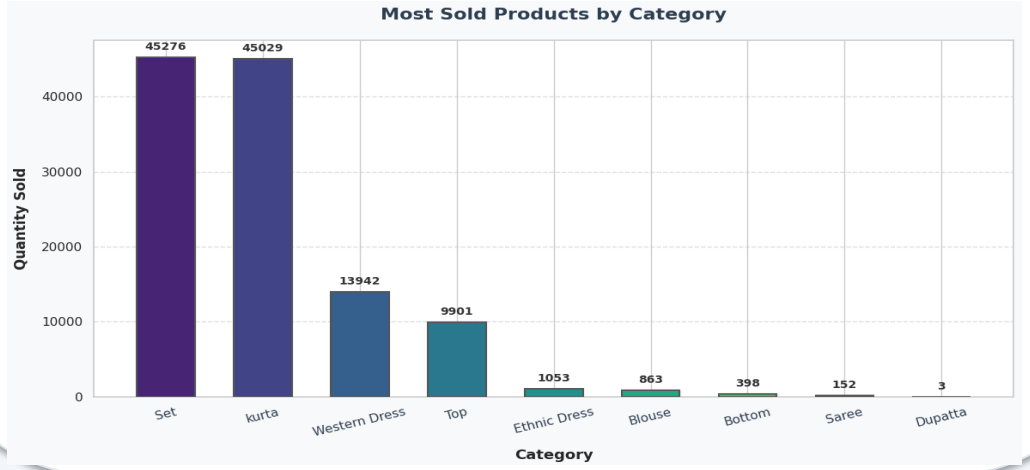
## Query output

| category | size | total_amount | rnk |
|---|---|---|---|
| Set | M | 7302732.75 | 1 |
| kurta | L | 3727685.84 | 1 |
| Western Dress | L | 2026955.69 | 1 |
| Top | M | 944188.71 | 1 |
| Ethnic Dress | XL | 135239.29 | 1 |
| Saree | Free | 123933.76 | 1 |
| Blouse | M | 80511.24 | 1 |
| Bottom | XL | 25879.63 | 1 |
| Dupatta | Free | 915.00 | 1 |

## Top category-size combinations




**Leading combination: Set / M — 7.30M in sales amount.**



# 04 | Top 10 SKUs by sales amount

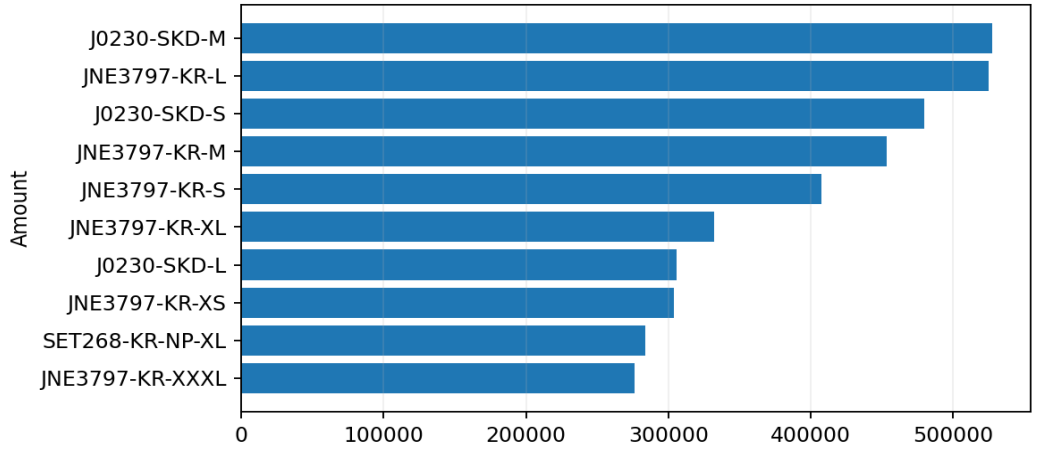
## SQL query

```sql
select sku,
       sum(amount) as total_amount,
       round(avg(amount), 2) as avg_amount
from amazon
group by sku
order by total_amount desc
limit 10;
```

## Query output

| sku | total_amount | avg_amount |
|---|---|---|
| J0230-SKD-M | 527699.20 | 1118.01 |
| JNE3797-KR-L | 524581.77 | 726.57 |
| J0230-SKD-S | 479937.14 | 1121.35 |
| JNE3797-KR-M | 453555.16 | 731.54 |
| JNE3797-KR-S | 407302.57 | 736.53 |
| JNE3797-KR-XL | 332155.24 | 744.74 |
| J0230-SKD-L | 305616.95 | 1111.33 |
| JNE3797-KR-XS | 303616.70 | 745.99 |
| SET268-KR-NP-XL | 284058.96 | 763.60 |
| JNE3797-KR-XXXL | 276375.80 | 738.97 |

## Top SKU revenue




**Top SKU: J0230-SKD-M — 527.7K sales amount.**

# 05 | Geographic sales leaders

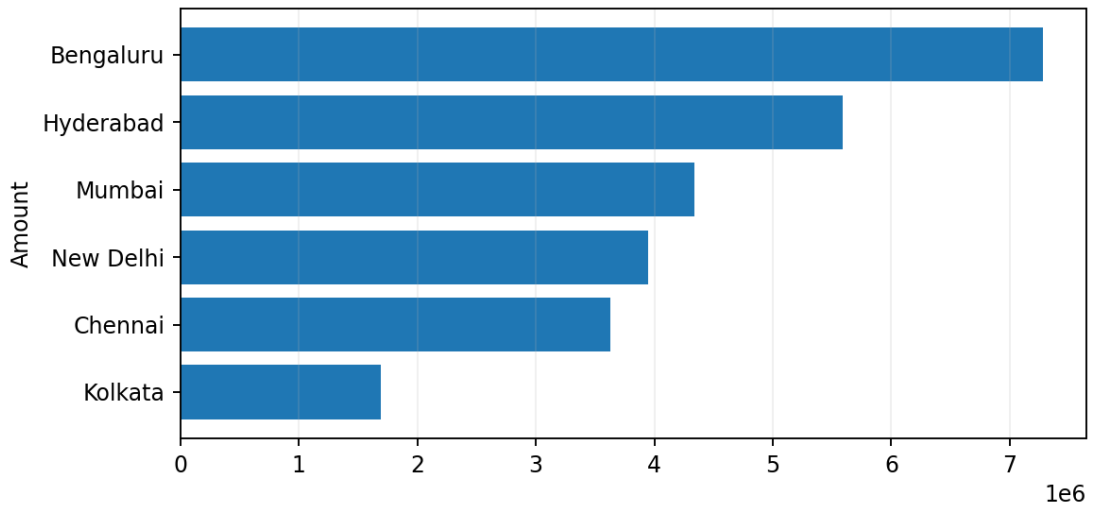
## SQL query

```sql
with temp as (
    select state, city,
           sum(amount) as total_amount,
           row_number() over (partition by state order by sum(amount) desc) as rnk
    from amazon
    group by state, city
)
select state, city, total_amount, rnk
from temp
where rnk < 2
order by total_amount desc limit 6 ;
```

## Query output

| state | city | total_amount | rnk |
|---|---|---|---|
| Karnataka | Bengaluru | 7279086.80 | 1 |
| Telangana | Hyderabad | 5586637.35 | 1 |
| Maharashtra | Mumbai | 4336947.07 | 1 |
| Delhi | New Delhi | 3951323.25 | 1 |
| Tamil Nadu | Chennai | 3630238.94 | 1 |
| West Bengal | Kolkata | 1692661.89 | 1 |

## Revenue by leading city/state result




**Bengaluru leads the displayed results at 7.28M, followed by Hyderabad at 5.59M.**

# 06 | City cancellation & return rates

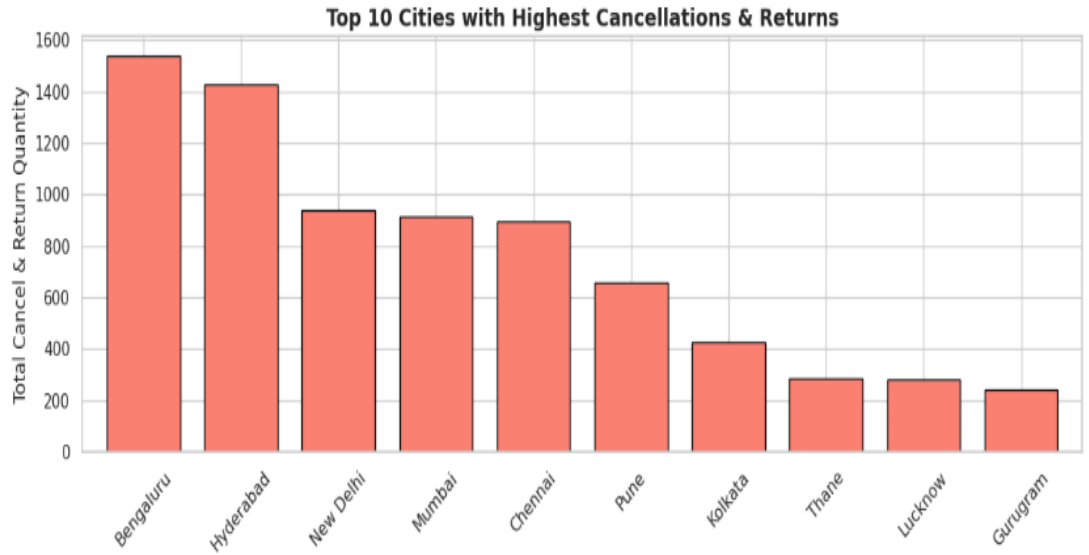
## SQL query

```
with rate as (select city,
       count(*) as total_orders,
       sum(status like 'Cancelled%') as cancelled,
       sum(status like '%Return%' ) as returned,
       sum(status like '%Delivered%') as delivered
from amazon
group by city
order by total_orders desc)
select city, total_orders , concat(round((cancelled/total_orders)*100,2),"%") as cancelled_rate,
concat(round((returned/total_orders)*100,2),"%") as retun_rate from rate  limit 6;
```

## Query output

| city | total_orders | cancelled_rate | retun_rate |
|---|---|---|---|
| Bengaluru | 11933 | 11.82% | 1.15% |
| Hyderabad | 9140 | 14.34% | 1.37% |
| Mumbai | 7192 | 12.00% | 0.78% |
| New Delhi | 6364 | 13.29% | 1.57% |
| Chennai | 6318 | 13.09% | 1.17% |
| Pune | 4651 | 12.79% | 1.40% |

## Cancellation rate by city




**Highest cancellation rate in the shown cities: Hyderabad (14.34%). New Delhi has the highest displayed return rate (1.57%).**

# 07 | Fulfilment performance comparison

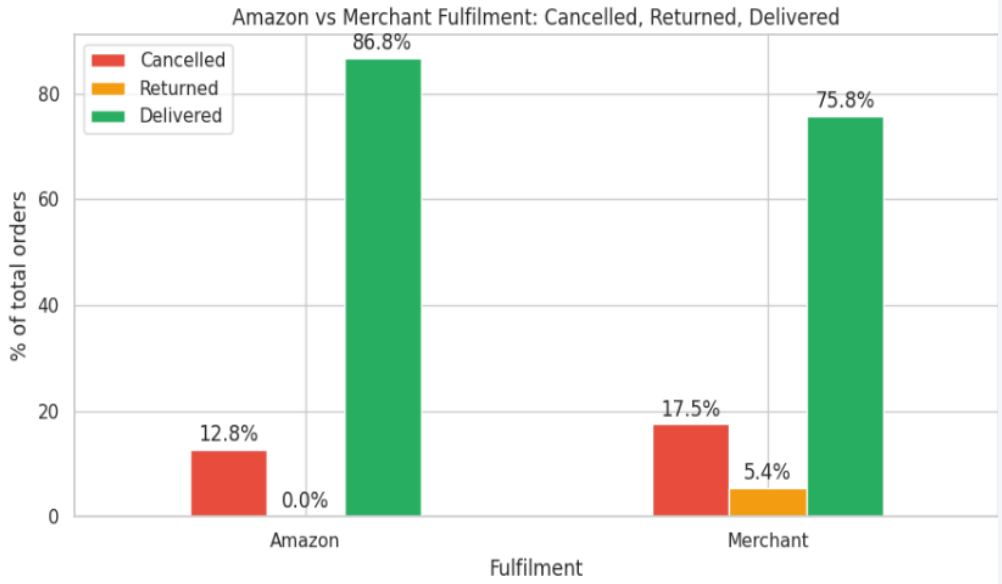
### SQL query screenshot

```sql
select fulfilment,
       count(*) as total_orders,
       sum(status like 'Cancelled%') as cancelled,
       sum(status like '%Return%') as returned,
       sum(status like '%Delivered%') as delivered,
       round(sum(status like 'Cancelled%') / count(*) * 100, 2) as cancel_pct,
       round(sum(status like '%Return%') / count(*) * 100, 2) as return_pct,
       round(sum(status like '%Delivered%') / count(*) * 100, 2) as delivered_pct
from amazon
group by fulfilment;
```

| fulfilment | total_orders | cancelled | returned | delivered | cancel_pct | return_pct | delivered_pct |
|---|---|---|---|---|---|---|---|
| Merchant | 39262 | 6858 | 2107 | 29768 | 17.47 | 5.37 | 75.82 |
| Amazon | 89676 | 11466 | 0 | 77795 | 12.79 | 0.00 | 86.75 |

### Key insight

Merchant cancels 4.68 points more than Amazon (17.47% vs 12.79%). Returns appear only for Merchant (5.37%); Amazon-fulfilled returns are not recorded, so its 0% is not a real rate.



# 08 | Cancellation patterns by category

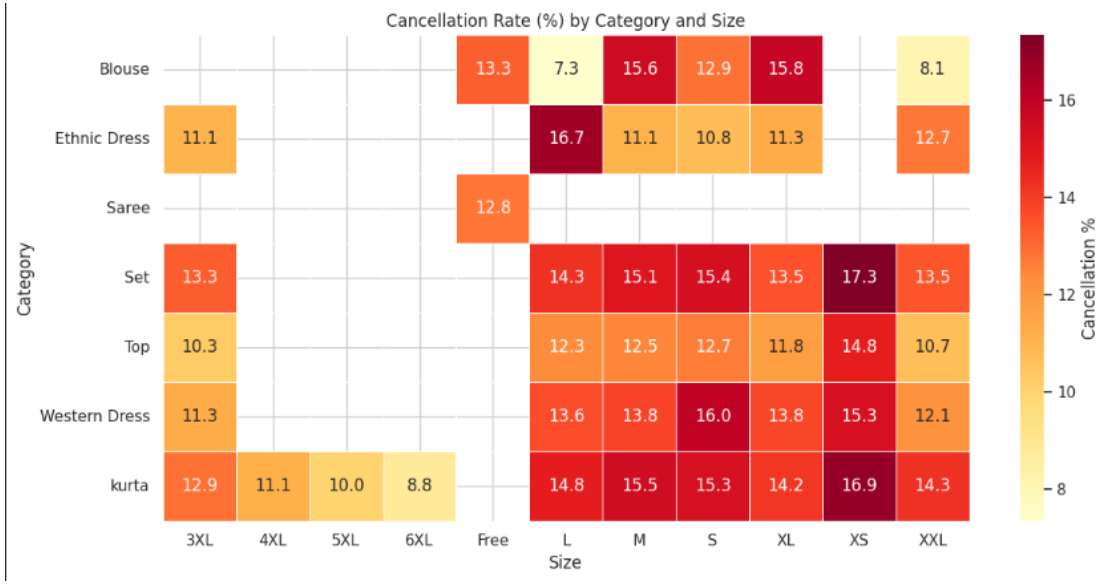
## SQL query

```sql
select category,
       count(*) as total_orders,
       sum(status like 'Cancelled%') as cancelled,
       round(sum(status like 'Cancelled%') / count(*) * 100, 2) as cancel_pct
from amazon
group by category
having count(*) >= 100
order by cancel_pct desc;
```

## Result sample

| category | total_orders | cancelled | cancel_pct |
|---|---|---|---|
| Set | 50270 | 7336 | 14.59 |
| kurta | 49857 | 7248 | 14.54 |
| Western Dress | 15499 | 2122 | 13.69 |
| Bottom | 440 | 60 | 13.64 |
| Saree | 164 | 21 | 12.80 |
| Blouse | 926 | 116 | 12.53 |
| Ethnic Dress | 1159 | 145 | 12.51 |
| Top | 10620 | 1276 | 12.02 |

## Selected category-size cancellation rates




**Highest in this visible sample: Bottom / XXL at 18.92%, followed by Blouse / XS at 17.86%.**

These are sample rows, not necessarily the highest across the full dataset; compare groups with adequate order volume.

# 09 | Color-wise sales performance

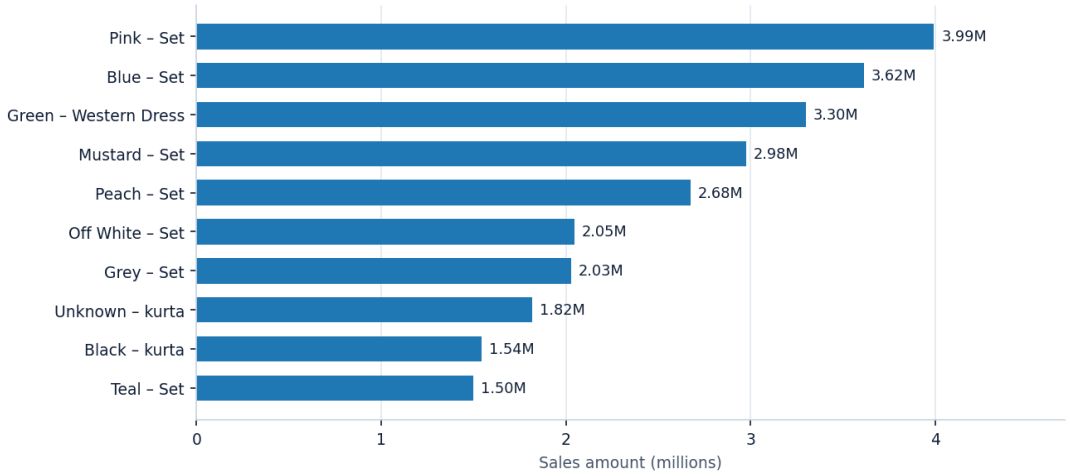
## SQL query

```
with color as (SELECT color, category,
        SUM(amount) AS total_amount,
        RANK() OVER (PARTITION BY color ORDER BY SUM(amount) DESC) AS top_selling
FROM amazon
where status <>"cancelled"
GROUP BY color, category)
SELECT color, category,total_amount
        from color
        where top_selling = "1" order by total_amount desc limit 10;
```

## Result screenshots

| color | category | total_amount |
|---|---|---|
| Pink | Set | 3994438.00 |
| Blue | Set | 3615494.00 |
| Green | Western Dress | 3298439.00 |
| Mustard | Set | 2976233.00 |
| Peach | Set | 2676277.00 |
| Off White | Set | 2045934.00 |
| Grey | Set | 2028370.00 |
| Unknown | kurta | 1817351.00 |
| Black | kurta | 1543708.00 |
| Teal | Set | 1499074.00 |

## Top category within each color




**Pink–Set leads at 3.99M, then Blue–Set (3.62M) and Green–Western Dress (3.30M). Each bar is a color's best-selling category, not its total.**

PROJECT WRAP-UP

# Executive Summary

A concise view of the key findings from the SQL analysis in this report

## 01 Revenue trend

Revenue amount peaks in April, then declines by 9.1% in May and 10.7% in June.

March contains only one day of data.

## 02 Customer segment

Average order value is ₹768 for B2B vs ₹695 for B2C. B2B volume is small (728 orders), so treat this as indicative.

## 03 Product performance

Set / M is the leading category-size combination at 7.30M. Top SKU in the displayed analysis is J0230-SKD-M at 527.7K.

## 04 Geography

Bengaluru leads the displayed city results at 7.28M, followed by Hyderabad at 5.59M.

## 05 Operations

Merchant cancellation rate is 17.47% vs 12.79% for Amazon fulfilment, a 4.68 percentage-point gap.

## 06 Data caveat

Some amount totals include cancelled orders. Return status is only recorded for Merchant-fulfilled orders.

## How to read these findings

These are findings from the displayed query outputs; revenue figures should not be described as net revenue unless cancelled orders are excluded consistently.

ANALYTICAL APPROACH

# Methodology & Limitations

Transparency about the dataset and interpretation boundaries

## Analysis covered

- Monthly revenue amount trend
- B2B vs B2C average order value and basket size
- Category-size and SKU performance
- City-level sales and cancellation/return patterns
- Fulfilment comparison and colour/category analysis

## Data preparation documented in notebook

- Standardised column names and city/state text
- Consolidated order-status labels
- Added an Amount_Fill field for missing amounts
- Filled missing courier status as Cancelled
- Mapped missing colour values from style where available

## Important limitations

- Dataset covers 31 Mar–29 Jun 2022; March is only one day.
- Some reported amount totals include cancelled orders.
- Return status is recorded only for Merchant fulfilment.
- Small B2B sample means the segment gap is indicative.

## Next validation steps

- Recalculate sales with a clearly defined status filter.
- Check row count, distinct order count and duplicate handling.
- Validate category/city rankings using adequate order volume.
- Document metric definitions alongside each chart.

### Project scope note

The PowerPoint presents SQL query screenshots and result interpretations; the supplied notebooks document data preparation and Python-based exploratory visualisations.